

Nadejda V. Drenska

Curriculum Vitae

Department of Mathematics, Louisiana State University
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Research Interests

Broad: data science, nonlinear analysis, PDEs, repeated two-person games, graph theory, applications in computer science, financial mathematics, biomedical applications
Specific: semi-supervised learning, online machine learning problems from prediction with expert advice, viscosity solutions of PDEs, optimal control theory, body composition analysis, investment algorithms

Positions Held

Assistant Professor at the Department of Mathematics, Louisiana State University	2023-present
Rufus Isaacs Postdoctoral Fellow at Applied Mathematics and Statistics Department, Johns Hopkins University	2021-2023
MCFAM Postdoctoral Associate at the School of Mathematics, University of Minnesota, Twin Cities	2018-2021

Education

New York University – Courant Institute of Mathematical Sciences	2017
Ph.D. in Mathematics	
Thesis advisor Professor Robert V. Kohn,	
Thesis topic: A PDE Approach to a Prediction Problem Involving Randomized Strategies	
Brown University	2012
B. Sc. in Mathematics with Honors and B. Sc. in Applied Mathematics with Honors, <i>magna cum laude</i>	
Applied mathematics thesis advisor Bjorn Sandstede,	
Thesis topic: Numerical Approximation of Spectra for Localized Oscillatory Structures	
Mathematics thesis advisor Jill Pipher,	
Thesis topic: Representation of Periodic Data with Fourier Methods and Wavelets	

Grant History

National Science Foundation: Machine Learning, Nonlinear PDEs, and Biomedical Applications	2024
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Publications and Manuscripts

16) Nadejda Drenska. **On the Relationship between Prediction with Expert Advice and the Hedge Allocation Problem** (*in preparation*)
15) Nadejda Drenska, Wonjun Lee. **Continuous Techniques for Adaboost and Drifting Games** (*in preparation*)

- 14) Jeff Calder, Nadejda Drenska, Hao Lan. **Explicit Solutions to Prediction with Expert Advice Involving Randomized Strategies.** (*in preparation*)
- 13) Nadejda Drenska, Matthew Lemoine, Gowri Priya Sunkara, Jonathan Wang, Sri Devrakonda, and Steven Heymsfield. **Target-Aware Variational p-Laplacian Graph Regression for Non-Invasive Body Composition Estimation** (*in preparation*)
- 12) Christian Ennis, Nadejda Drenska, Cayden Huffman, Devin Johnson, Jacob Johnston, Kennedy Jeffery, Faris Khattak, Haoxiang Lin, Bayla Patin. **Performance in a College Algebra Course with Corequisites: Evidence for Early Intervention and Consistent Patterns** (*submitted in PRIMUS*)
- 11) Sarah Bodenstein, Dow Draper, Mark C. Herpin, Austin C. Louque, Sanjida Nupur, Maxwell T. Shroyer, Alex Mensen Johnson, Gowri Priya Sunkara, Shalini Shalini, Nadejda Drenska, Peter R. Wolenski, Elizabeth M. Robinson. **Applying Machine Learning Tools to Advance Quality Control in Oyster Nurseries** (*Aquacultural Engineering*)
- 10) Gyaneshwar Agrahari, Kiran Bist, Monika Pandey, Jackson Knox, Zachary James, Steven Heymsfield, Sophia Ramirez, Peter Wolenski, Nadejda Drenska. **Predicting Anthropometric Body Composition Variables using 3D Optical Imaging and Machine Learning** *Frontiers in Bioinformatics*
- 9) N. Drenska **Games on deBruijn Graphs and Cycle Means** *Discrete Applied Mathematics*, <https://doi.org/10.1016/j.dam.2025.12.060>
- 8) A. Bertozzi, N. Drenska, J. Latz, and M. Thorpe. **Partial Differential Equations in Data Science.** *Philosophical Transactions of the Royal Society A*, Volume 383, issue 2298, <https://doi.org/10.1098/rsta.2024.0249>
- 7) D. Mosaphir, J. Calder, and N. Drenska. **Numerical Solution of a PDE Arising from Prediction with Expert Advice.** *The European Journal of Applied Mathematics*, <https://doi.org/10.1017/S0956792525000075>
- 6) J. Calder and N. Drenska. **Consistency of Semi-Supervised Learning, Stochastic Tug-of-War Games, and the p-Laplacian.** *Active Particles, Volume 4. Modeling and Simulation in Science, Engineering and Technology*. 2024. https://doi.org/10.1007/978-3-031-73423-6_1
- 5) N. Drenska and J. Calder. **Online Prediction with History-Dependent Experts: The General Case.** *Communications on Pure and Applied Mathematics (CPAM)*, 2022, <https://doi.org/10.1002/cpa.22049>
- 4) N. Drenska and R. V. Kohn. **A PDE Approach to the Prediction of a Binary Sequence with Advice from Two History-Dependent Experts.** *Communications on Pure and Applied Mathematics (CPAM)*, 2022 <https://doi.org/10.1002/cpa.22071>
- 3) J. Calder and N. Drenska. **Asymptotically Optimal Strategies for Online Prediction with History-Dependent Experts.** *Journal of Fourier Analysis and Applications*, **27**, article 20, 2020, <https://doi.org/10.1007/s00041-021-09815-4>
- 2) N. Drenska and R.V. Kohn. **Prediction with Expert Advice: a PDE Perspective.** *Journal of Nonlinear Science*, **30(1)**: 137-173, 2020, <https://doi.org/10.1007/s00332-019-09570-3>
- 1) N. Drenska. **A PDE Approach to a Prediction Problem Involving Randomized Strategies.** PhD thesis, New York University, 2017

Select Talks

Body Composition Analysis: Insights Through Regression and Machine Learning	2025
Keynote Speaker, 8th International Science Conference - Federal University Oye-Ekiti, Nigeria	
A Journey in Machine Learning	2025
Math Circle, LSU	
Body Composition: Insights Through Regression and Machine Learning	
Math Club, LSU	2024
SIAM Annual Meeting	2024

Nadia Drenska's Machine Learning Journey

The Johns Hopkins University	2024
Louisiana State University	2023

Semi-Supervised Learning with the p -Laplacian in Geometric Methods in Machine Learning and Data Analysis

Numerical PDEs: Analysis, Algorithms, and Data Challenges, ICERM, March 2024	2024
International Congress on Industrial and Applied Mathematics	2023
Optimal Investment: Robo-Advising Under Small Changes of Risk Aversion	
Joint Mathematics Meetings	

A PDE Interpretation of Prediction with Expert Advice

University of Vermont	2023
University of North Carolina, Charlotte	2023
Louisiana State University	2023
University of Maryland, Baltimore County	2023
North Carolina State University	2023
University of Rhode Island	2023
NJIT	2022
Johns Hopkins Applied Mathematics and Statistics Colloquium	2021
JMU Artificial Intelligence and Machine Learning Seminar Series	2021
WPI Colloquium	2021
Joint Mathematics Meetings	2021
OneWorld Machine Learning	2020
LMS-Bath Symposium	2020

Two PDE Approaches to A Problem from Prediction with Expert Advice

IPAM, UCLA	2020
Analysis and Applied Mathematics Seminar, UIC	2020

PDE Approaches to Two Problems from Prediction with Expert Advice

Applied Interdisciplinary Mathematics Seminar, UMichigan	2019
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A PDE Approach to Some Randomised-Strategy Two-Player Games

IMA Data Science Seminar, UMN	2018
Materials Working Groups, NYU	2016

A PDE Approach to Prediction with Expert Advice

WPI STEM Faculty Launch, WPI	2016
RPI Applied Math Days, RPI	2016
SIAM Conference on Analysis of PDEs, Scottsdale AZ (awarded SIAM Student Travel Award)	2015
Materials Working Group, NYU	2015

Teaching Experience

Department of Mathematics, Louisiana State University

Instructor for 4997 (VIR Machine Learning)	2026
Instructor for 2057 (Multidimensional Calculus)	2026
Instructor for 4020 (Machine Learning Capstone)	2025
Instructor for 4997 (VIR Machine Learning)	2025
Instructor for 4020 (Machine Learning Capstone)	2024
Instructor for 4997 (VIR Machine Learning)	2024
Instructor for 2057 (Multidimensional Calculus)	2024

Instructor for 4020 (Machine Learning Capstone)	2023
Applied Mathematics and Statistics Department, Johns Hopkins University	
Instructor for Probability and Statistics for the Life Sciences	2021-2023
Instructor for and developer of Freshman Experience Course ‘Mathematics in Baseball’	2021
University of Minnesota	2018-2021
Instructor for Multivariable Calculus, PDEs I and II	
Instructor and course supervisor for 13 Multivariable Calculus sections	2018
Courant Institute of Mathematical Sciences, NYU	2014, 2015
Teaching Assistant for Calculus I, PDEs, and ODEs	
Mathematics Department, Brown University	2009, 2010, 2012
Teaching Assistant and/or grader for Analysis, ODEs, PDEs, Multivariable Calculus	
Division of Applied Mathematics, Brown University	2011
Teaching Assistant for Methods of Applied Mathematics I, Methods of Applied Mathematics II	
Math Resource Center, Brown University	2009
Tutor for calculus, linear algebra, and methods of applied mathematics (differential equations)	

Teaching High School Students

Organizer and speaker at LSU Math Circle’s Machine Learning Day	2025
Guest lecturer for the LSU Math Circle	2024
Instructor and co-organizer for Machine Learning Virtual Summer Camp for high school students	2020

Awards and Recognition

Moses A. Greenfield Research Award for Outstanding Interdisciplinary studies, The Courant Institute, NYU	2016
Rohn Truell Prize to an outstanding undergraduate student in the Division of Applied Mathematics, Brown University	2012
Sarah Dyer Barnes Scholarship – Brown University	2011-2012
Henry Parker Manning Prize Examination – 1 st prize	2011
Graduated (high school) with Recognition for Outstanding Achievements in the Areas of Mathematics and Physics	2007
National Diploma for Outstanding Achievements from the Minister of Education of Bulgaria	2007
Member of the Bulgarian Extended National Team for the International Mathematics Olympiad	2007
Member of the Bulgarian Extended National Team for the Balkan Mathematics Olympiad	2005
1 st and 2 nd prizes at National Physics Competitions in Bulgaria	2005-2006

Service

LSU Math Club Adviser	2024-present
Member of Undergraduate Awards Committee, Mathematics Department, LSU	2024-2026
Member of Undergraduate Awards Committee, LSU College of Science	2024-2026
Helped Establish the New Financial Mathematics Concentration, LSU	2025
Guest Editor of Philosophical Transactions of the Royal Society A: ‘PDEs in Data Science’	2024-2025
Elected Postdoc Representative, Applied Mathematics and Statistics, Johns Hopkins University	2021-present

Co-organized an IMA workshop ‘Optimal Control, Optimal Transport, and Data Science’	2020
with Jeff Calder, Dejan Slepcev, and Chai Wu	
Co-organized a minisymposium ‘Partial Differential Equations in Machine Learning and Data Science’ with Jeff Calder at the SIAM Conference on Analysis of PDEs	2017
President of The Courant Student Organization	2014-2015
President of The Department Undergraduate Group of Applied Mathematics	2011-2012